MASTERING

MSAT

A three-band analogue saturation unit in the spirit of a mid-70s British design, with a transient shaper and stereo width on every band.



Overview

MSAT splits the signal into LOW, MID and HIGH bands and gives each band the same three tools: a transient shaper, a warm discrete-transistor saturation and stereo width. The sound is designed from scratch.

The split is a Linkwitz-Riley crossover at 24 dB per octave, with the low band phase-aligned to the upper split, so the three bands sum back flat: with DRIVE at 0 the whole unit measures flat within about 0.05 dB. Each band's frequency range is shown beside its name.

DRIVE is level-compensated, so it changes the colour rather than the level: warm 2nd and 3rd harmonics that grow as you push it, peaks softened. The transient shaper is level-independent, with its envelope times tuned per band (slower in the lows, faster in the highs). A mix tool, its output stage passes straight through (no clip).

Reach for it for

- Analogue warmth on a mix bus or master, band by band
- Punchier drums, or tamed snap, without a compressor
- Mono lows and wider highs in one pass
- Thickening a bass or a vocal in just the part of the spectrum that needs it

Controls

The header

PRESETS	Opens the full preset list, with Reset all to defaults at the end.
◀ / ▶	Step to the previous or next preset (wrapping round).
Preset display	Shows the preset name. Click it for the preset list too.
VFD / MATRIX	The panel's look: VFD (dark green, light green and black, the default) or MATRIX (battleship grey, elephant's breath print and mustard amber, as on KRT81). The look only: the sound is the same.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window. New instances open in medium, on Init.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

Knobs: drag up or down (Shift for fine moves) or use the mouse wheel; double-click resets a knob to its default. The label under a knob turns into its value while you touch it.

INPUT and BAND LEVEL

INPUT	-12 to +12 dB (0 dB): the level into the unit, and so into the saturation.
BAND LEVEL	Each band's output level, left and right, from -48 to 0 dBFS, and beside it the band's TRANSIENT bar: the shaper's live gain, from -15 dB (amber, left) to +15 dB (green, right).

The crossovers

LOW / MID	50 Hz - 1 kHz (250 Hz).
MID / HIGH	1 - 12 kHz (4 kHz). It never sits below 1.5 times the LOW / MID point.

Both glide as you turn them, so they don't zip.

Each band: LOW, MID, HIGH

S / M	SOLO and MUTE. With any band soloed, only the soloed bands play (solo two to hear them together); M takes a band out. Both fade over 4 ms, so they don't click, and the band meters keep showing every band.
HEAT	A lamp that glows as the band warms up (see DRIFT).
ATTACK	-15 to +15 dB (0): lifts or softens the start of each hit. It leaves the tails alone.
SUSTAIN	-15 to +15 dB (0): lifts or tucks the tail of each sound.
DRIVE	0 to 24 dB (6 dB): how hard the band is driven into the saturation, level-compensated. At 6 dB it adds a gentle warmth (about 1% THD on a -12 dBFS tone), at 12 dB a clear colour (about 4%), and at 24 dB it is thick and dense (around 30%).
MIX	0 - 100% (100%): the saturated band against the clean band, in parallel and in phase.
WIDTH	0 - 200% (100%): 0% is mono, 100% as it was, 200% wide (the side doubled, the level kept steady).



OUTPUT and the bottom strip

OUTPUT	-12 to +12 dB (0 dB).
DRIFT	0 - 100% (50%, the standard amount; 0 = none): heat and bias. See DRIFT below.
IN / OUT	The input and output meters, with the STEREO / MONO state beside them.

DRIFT

Heat and bias. Each band warms with what it is fed, rising over a couple of seconds and cooling over about six: as it warms, its bias leans further (more 2nd harmonic) and its drive edges up, by up to 1.5 dB. Each band's bias also wanders slowly on its own path. At 0 it all holds still.

Presets

Twelve presets, Init and eleven more. A preset sets the crossovers and the three bands; INPUT, OUTPUT, the look and DRIFT stay as they are. Every preset plays within about 2 dB of Init.

Init	Every band at DRIVE 6 dB, MIX 100%, no transient shaping, normal width.
Gentle Warmth	A little more drive, mixed in parallel, the highs a touch wider.
Punchy Drums	ATTACK up in every band and SUSTAIN down in the lows and mids, with drive: tighter, harder hits.
Tame the Snap	ATTACK down, most in the highs: softer, rounder transients.
Wide Air	The highs wide (150%) with a little sustain and drive.
Mono Lows	The lows below 150 Hz in mono, the highs a touch wider.
Vocal Presence	Drive and attack in the mids and highs, the lows tucked.
Bass Glue	The lows below 120 Hz driven hard and in mono, with sustain.

Room Lift	SUSTAIN up in every band and the upper bands wider: more room.
Snappy Mix Bus	Attack, drive and a little width across the bands, mixed in parallel.
Thick and Hot	Heavy drive in every band.
Clean Width	No saturation (MIX 0): width only, narrower lows and wider highs.
Reset all to defaults	(Last in the list.) Init, with INPUT, OUTPUT, the look and DRIFT back to their defaults too.

Tips

- On a mix bus, start at Gentle Warmth and bring MIX down until you only just hear it.
- Set the LOW / MID crossover between the kick and the bass and the drums' body, then shape each separately.
- For width without colour, use Clean Width: MIX at 0 bypasses the saturation.

`File: gfx-msat.jsfx`